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Curriculum Vitae

Haein Cho

Ph.D.

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Researcher in bio-compatible intelligent systems integrating neuromorphic devices, flexible electronics, and device–system integration. My work focuses on interfacing real-world biological and sensory signals with local hardware processing through integrated sensing, computation, and memory. Combining expertise in neuromorphic devices, skin-conformal electronics, and cleanroom-based nanofabrication, I develop scalable electronic platforms for human-machine interfaces and next-generation bioelectronic systems, with representative publications in *Nature Electronics*, *Nature Reviews Electrical Engineering*, and *npj Unconventional Computing*.

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1. Education & Research Experience

2025-2026	University of Virginia, USA **Postdoctoral Researcher**, Electrical and Computer Engineering Supervisor: Kyusang Lee
2024-2025	University of Virginia, USA **Visiting Scholar**, Electrical and Computer Engineering
2024	Korea University, Korea **Postdoctoral Researcher**, KU-KIST Graduate School of Converging Science and Technology
2018-2024	Korea University, Korea **Ph.D.**, KU-KIST Graduate School of Converging Science and Technology (Nano Convergence Research Group) Advisor: Gunuk Wang
2013-2018	Korea University, Korea **B.S**. in Chemical and Biological Engineering

2. Research Interests

- Bio-compatible intelligent systems for human-machine interfaces
- Flexible and soft electronics for biological signal interfaces
- Memristive devices for neuromorphic computing
- Temporal information processing for real-world sensory data
- Monolithic integration and scalable nanofabrication

3. Journal Publications

Selected First / Co-first Authorship

1. I. Lee[†], H. Shin[†], H. Cho[†], J.-G. Choi, G. Wang[*] and S. Park[*], "Skin-conformal electronics for intelligent gesture recognition", ***Nat. Rev. Elec. Eng.,*** 2, 736–754 (2025).

2. Y. Baek[†], B. Bae[†], H. Shin[†], C. Sonnadara[†], H. Cho[†], C.-Y. Lin[†], Y. Mu[†], C. Shen[*], S. Shah[*], G. Wang[*], and K. Lee[*], "Edge intelligence through in-sensor and near-sensor computing for the artificial intelligence of things", ***npj Uncon. Com***., 2, 25 (2025).

3. H. Cho[†], I. Lee[†], J. Jang[†], J.-H. Kim, H. Lee, S. Park[*] and G. Wang[*], "Real-time finger motion recognition using skin-conformable electronics", ***Nat. Electron.,*** 6, 619–629 (2023). ([†]The authors contributed equally to this work; *introduced in News and Views: Human-machine interfaces, Integrated devices that can recognize hand gestures*)
4. H. Cho, C.-W. Jeon, B.-D. On, I.-K. Park, S. Choi, J. Jang[*] and G. Wang[*], "Transparent and unipolar nonvolatile memory using two-dimensional vertically stacked layered double hydroxide", *Adv. Mater. Inter.,* 2001990 (2021). *(Front cover selected)*

Co-authorship papers

5. Y. Park, H. Cho, and G. Wang[*], "Multiple switching modes of NiOx memristor for memory-driven multifunctional device applications" *ACS Appl. Elec. Mater.,* 4, 7, 3739–3748 (2022)
6. S. Choi, G. S. Kim, J. Yang, H. Cho, C.-Y. Kang and G. Wang[*], "Controllable SiO_x Nanorod Memristive Neuron for Probabilistic Bayesian Inference" *Adv. Mater.,* 34, 2104598 (2022)
7. S. Ham, S. Choi, H. Cho, S.-I. Na and G Wang[*], "Photonic Organolead Halide Perovskite Artificial Synapse Capable of Accelerated Learning at Low Power Inspired by Dopamine-facilitated Synaptic Activity" *Adv. Funct. Mater.,* 29, 1806646 (2019). *(Back Cover Feature)*

4. Mentoring Experience

- Mentored undergraduate and junior researchers in device fabrication, electrical characterization, and conference preparation.
- Co-authored publications with former undergraduate mentees.

5. Awards

- Young Scientist Award in IC ME&D (the 34th International Conference on Molecular Electronics and Devices), POSTECH, Pohang, Korea, May 8-10, 2024
- 2023 KU Achievement Award presented by Korea University Graduate School, Korea University, Seoul, Korea, Feb. 21, 2024
- Best Oral Presentation Award in KPS (The Korean Physical Society), CECO, Changwon, Korea, Oct. 24-27, 2023
- Best Oral Presentation Award in NCC (Nano Convergence Conference), Konjiam Resort,

Gyeonggi, Korea, Jan. 26-27, 2023

- Best Poster Presentation Award in KU-KIST symposium, Korea Institute of Science and Technology, Seoul, Korea, Dec. 17, 2021
- Best Oral Presentation Award in MRS-K (Materials Research Society of Korea), Sol Beach Hotel & Resort, Samcheok, Korea, Oct.30-Nov.1, 2019
- Best Poster Presentation Award in ICAMD (International Conference on Advanced Materials and Devices), Ramada Plaza Jeju Hotel, Jeju, Korea, Dec.10-13, 2019

6. Grants

- 2021 National R&D Real Challenge Program by KIRD (Korea Institute of Human Resource Development in Science and Technology)

7. Technical Skills

- End-to-end fabrication and characterization of neuromorphic hardware systems
- Thin-film deposition and micro/nano-patterning (sputtering, ALD, EBL, photolithography)
- Electrical measurement and reliability analysis (Keithley systems)
- Flexible device fabrication and surface characterization (PI/parylene, AFM, SEM)
- 3D device integration and BEOL-compatible process design for scalable hardware implementation
- Device-to-system integration for sensing-processing hardware platforms

8. References

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